Finite Automata

- Lex turns its input program into a lexical analyzer
- How?
  - By using a concept called finite automata
- **Finite automata are graphs, like transition diagrams, with a few differences**

1. Finite automata are ***recognizers;*** they simply say "yes" or "no" about each possible input string

2. Finite automata come in two types:
   a) ***Nondeterministic finite automata*** **(NFA**) have no restrictions on the labels of their edges. A symbol can label several edges out of the same state, and ε, the empty string, is a possible label.
   b) ***Deterministic finite automata*** **(DFA**) have, for each state, and for each symbol of its input alphabet exactly one edge with that symbol leaving that state.

- Both deterministic and nondeterministic finite automata are capable of recognizing the same languages which are called *regular languages*

Nondeterministic Finite Automata (NFA)

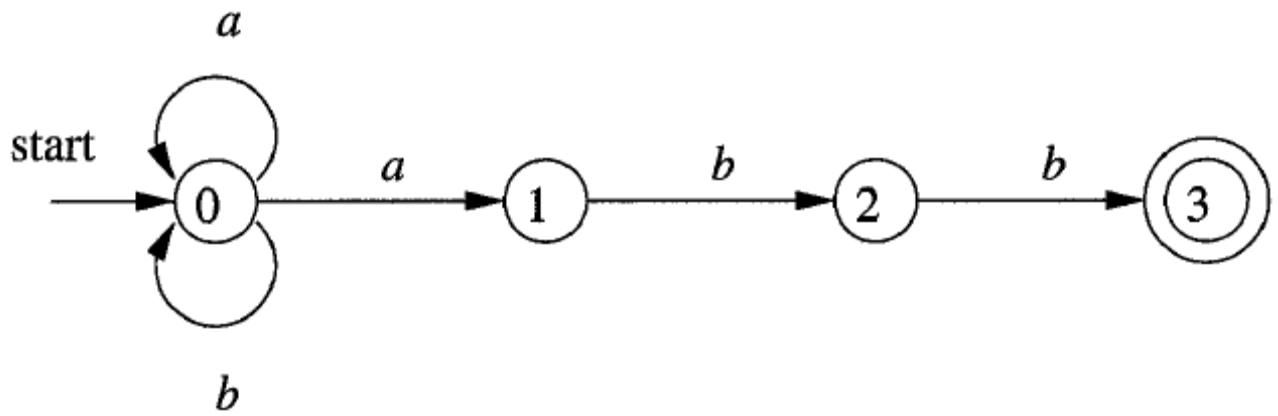
- NFA consists of
  1. A finite set of states S
  2. A set of input symbols Σ, the *input alphabet.* We assume that ε, which stands for the empty string, is never a member of Σ
  3. A *transition function* that gives, for each state, and for each symbol in $\Sigma \cup \{\varepsilon\}$ a set of *next states*
  4. A state s_0 from S that is distinguished as the *start state* (or *initial state)*
  5. A set of states F, a subset of S, that is distinguished as the *accepting states* (or *final states)*

- We can represent either an NFA or DFA by a *transition graph,* where the nodes are states and the labeled edges represent the transition function

- There is an edge labeled *a* from state s to state *t* if and only if *t* is one of the next states for state s and input *a*

- This graph is like a transition diagram, except:
  a) The same symbol can label edges from one state to several different states

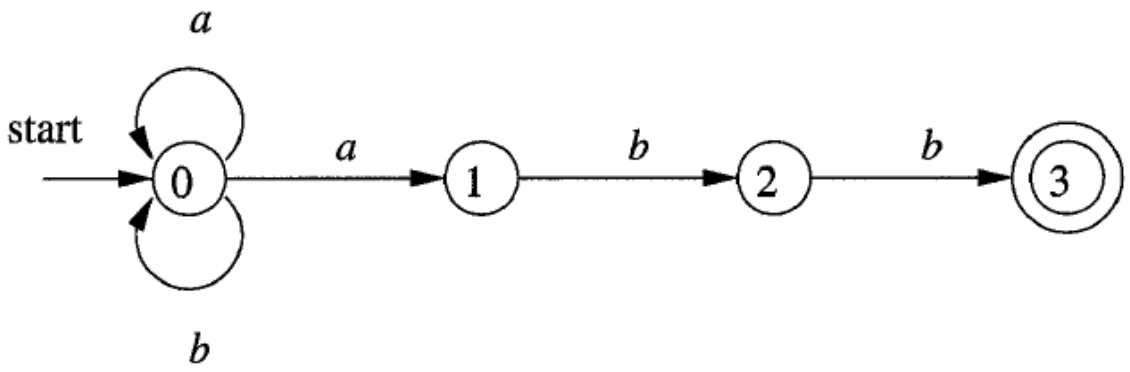
  b) An edge may be labeled by ε, the empty string, instead of, or in addition to, symbols from the input alphabet



NFA for the regular expression **(a|b)* abb**

Transition tables

- Represent an NFA by a **transition table**
  - Rows – states
  - Columns – input symbols and ε
- The entry for a given state and input is the value of the transition function applied to those arguments
- If the transition function has no information about that state-input pair, we put ϕ in the table for the pair



NFA for the regular expression **(a|b)* abb**

STATE	a	b	ϵ
0	$\{0,1\}$	$\{0\}$	$\emptyset$
1	$\emptyset$	$\{2\}$	$\emptyset$
2	$\emptyset$	$\{3\}$	$\emptyset$
3	$\emptyset$	$\emptyset$	$\emptyset$

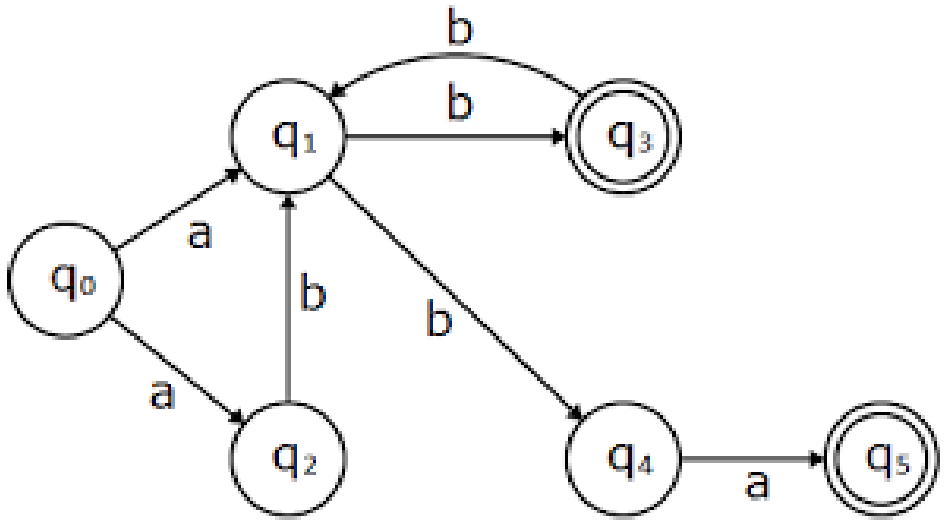
Transition table for the NFA (shown)

- Advantage of transition table
  - We can easily find the transitions on a given state and input

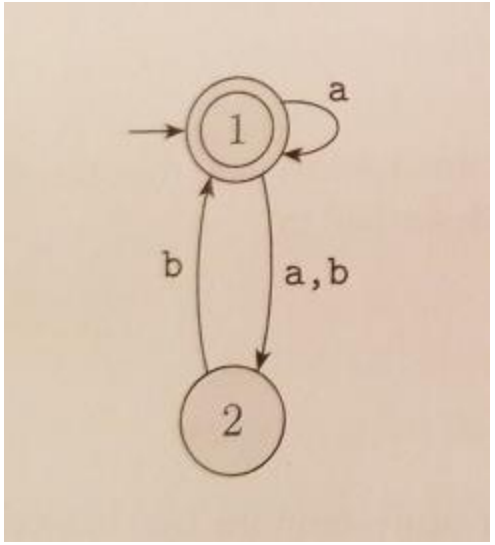
- Disadvantage of transition table
  - Takes a lot of space, when the input alphabet is large, yet most states do not have any moves on most of the input symbols

Exercise

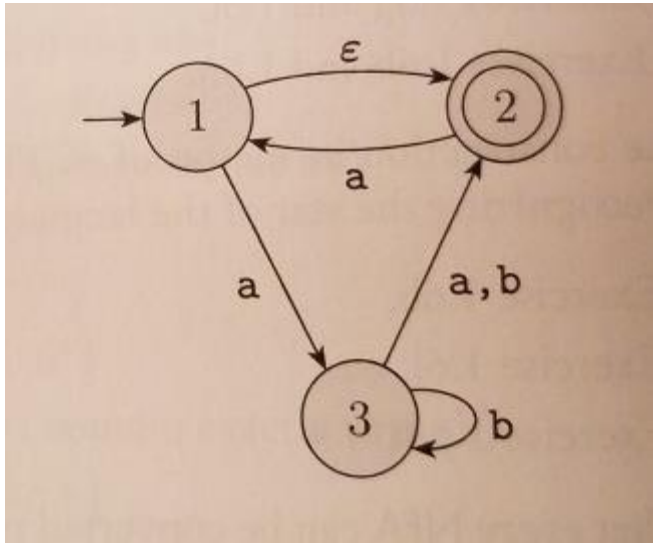
Draw the transition table for the NFA.



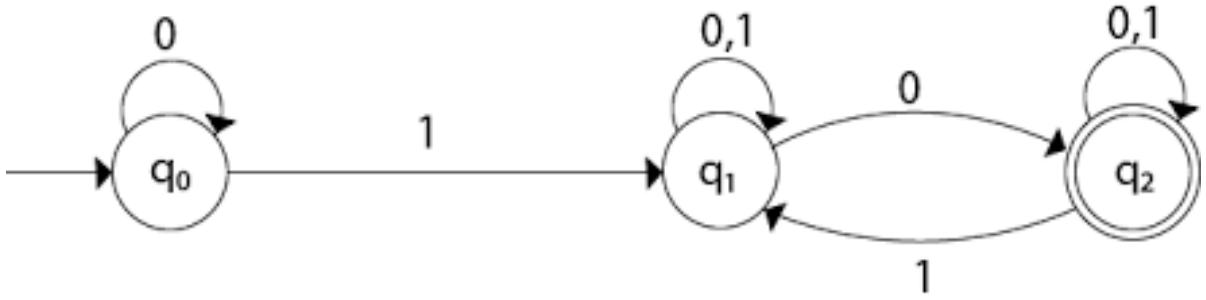
Draw the transition table for the NFA.



Draw the transition table for the NFA.



Draw the transition table for the NFA.

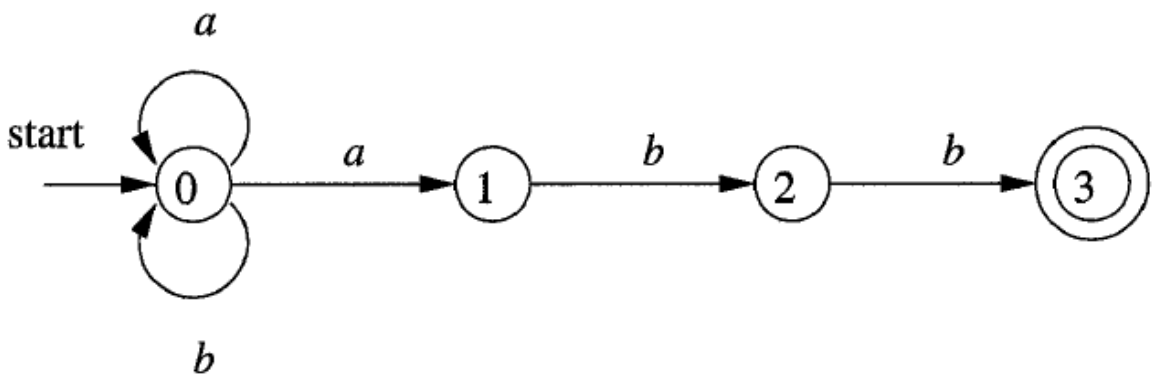


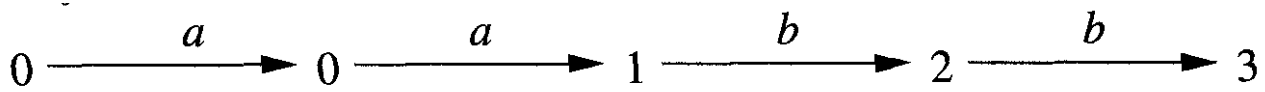
Acceptance of Input strings by automata

- An NFA *accepts* input string *x* if and only if there is some path in the transition graph from the start state to one of the accepting states, such that the symbols along the path spell out *x*

- Note that ε labels along the path are ignored, since the empty string does not contribute to the string constructed along the path

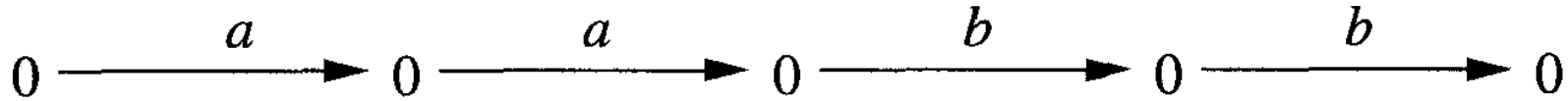
Example: The string "*aabb*" is accepted by the NFA





Several paths labeled by the same string may lead to different states

Another path from state 0 labeled by the string “*aabb*”



This path leads to state 0, which is not accepting

- NFA accepts a string as long as *some* path labeled by that string leads from the start state to an accepting state

- The existence of other paths leading to a non-accepting state is irrelevant

- The *language defined* (or *accepted)* by an NFA is the set of strings labeling some path from the start to an accepting state.

Reference Book

A. V. Aho, M. S. Lam, R. Sethi, and J. D. Ullman, *Compilers: Principles, Techniques & Tools*(2nd Edition), Pearson Education India , 2013